



LOWER TRUNK

The 12 thoracic vertebrae, 5 lumbar vertebrae, sacrum, ribs, and pelvic bones along with associated muscles and ligaments make up the flexible framework of the trunk. The vertebrae and the other bones, muscles, and ligaments work together to support and move the trunk. The vertebral bodies (the oval-shaped bones) of the trunk are connected by posterior and anterior ligaments along with other ligaments that connect each spinous and transverse (lateral bony protuberance) process to its corresponding part on the adjacent vertebrae. In addition, each vertebra is separated by an intervertebral disc. Compression of the vertebrae on the discs allows the trunk to move forward, backward, and sideways, with the amount of movement limited in part by the vertebral facets.

The trunk movements are flexion (moving the chest and thighs toward each other), extension (moving the chest and thighs away from each other), hyperextension (arching the trunk back beyond an erect position), lateral flexion (bending to the side to move the shoulder toward the same hip), lateral extension (bending to increase the distance between the shoulder and the same hip), and rotation.

Because many of the muscles in the trunk occur in right and left pairings, a muscle in the trunk can be involved in lateral flexion, lateral extension, and rotation. For example, the right external oblique and internal oblique abdominal muscles help perform right lateral flexion, and the left external oblique and internal oblique abdominal muscles help perform right lateral extension. Several of the muscles involved in movements of the lower trunk run between the pelvic bones and either the spinal column or rib cage.

The external oblique, internal oblique, and rectus abdominis of the abdomen (figure 5.1) and quadratus lumborum (figure 5.2a) flex the trunk by pulling the

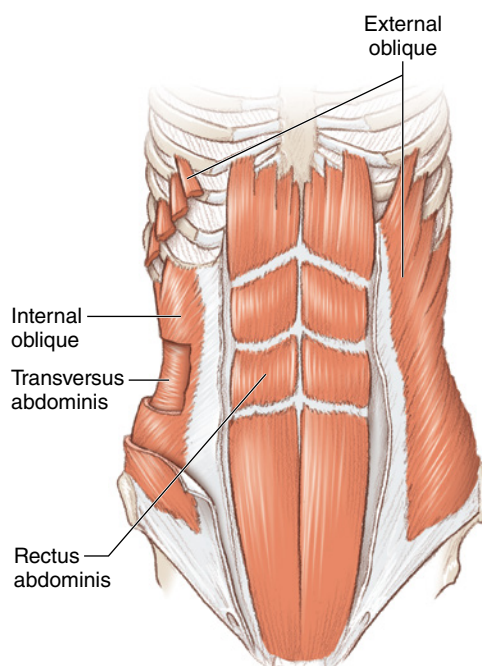


FIGURE 5.1 Abdominal muscles.

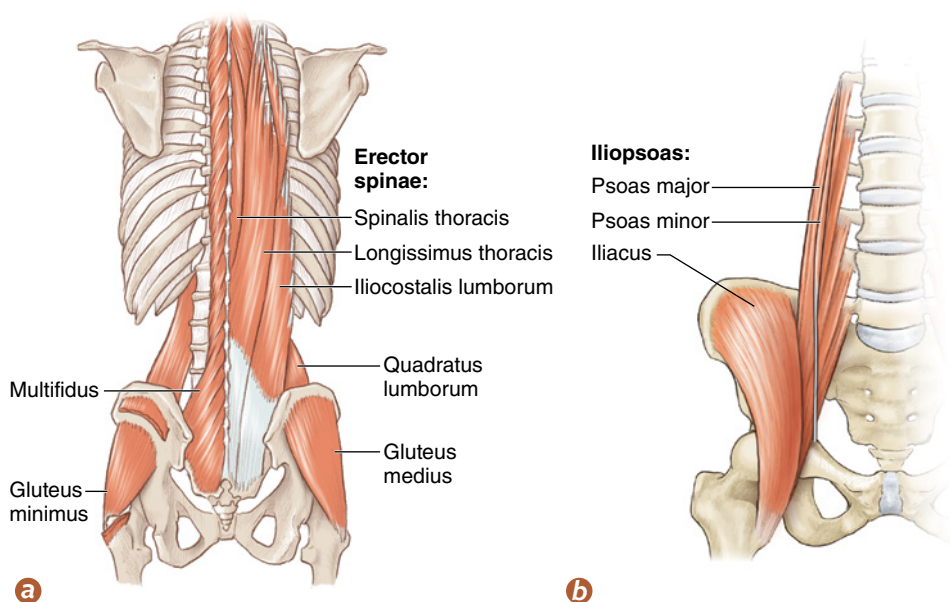


FIGURE 5.2 Core muscles: (a) posterior; (b) anterior.

rib cage toward the pelvis. The iliacus (figure 5.2b), a trunk flexor, pulls the femur (thigh bone) toward the pelvis. The psoas major, another trunk flexor, pulls the spinal column toward the femur. The prime trunk extensors (iliocostalis lumborum, longissimus thoracis, and spinalis thoracis) are collectively called the erector spinae. The iliocostalis lumborum runs between the posterior pelvis and posterior spinal column, while the longissimus thoracis and spinalis thoracis run along the posterior spinal column and help the individual vertebrae in the spinal column work together as a single unit. The interspinales, intertransversarii, multifidus, and rotatores run between individual vertebrae and cause large movements by making small changes between individual pairs or groups of vertebrae.

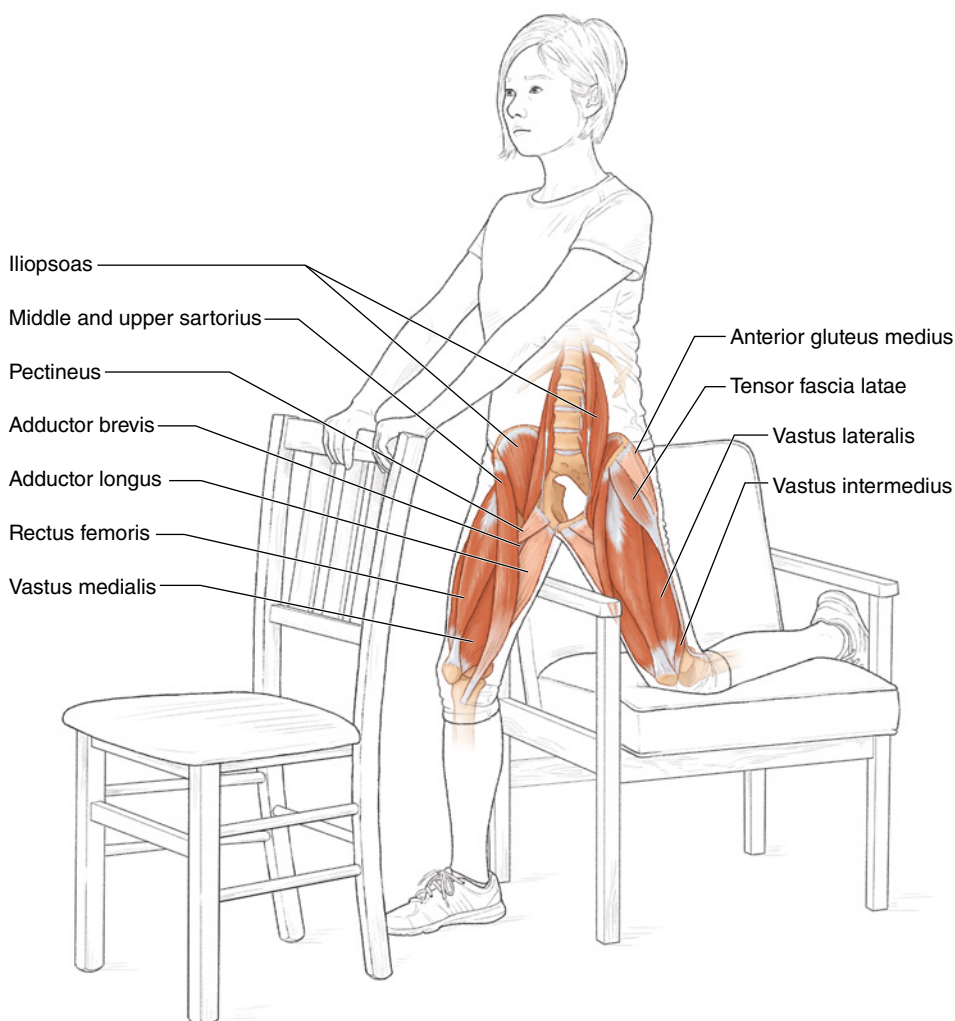
The ability to move the trunk is limited by the strength of the contracting muscles, the stiffness of the opposing ligaments, the stiffness of the non-contracting muscles, the alignment of the vertebral bodies with the adjacent vertebrae, the compressibility of the intervertebral discs, and the contact between body parts. For example, trunk flexion is limited by the stiffness of the posterior trunk muscles, the stiffness of the posterior trunk ligaments, the strength of the anterior trunk muscles, the alignment of the vertebral bodies with the adjacent vertebrae, the compressibility of the anterior portions of the intervertebral discs, the contact of the chin or rib cage with the legs, and the abdominal fat mass. Similarly, trunk extension is controlled by the stiffness of the anterior trunk muscles, the stiffness of the anterior trunk ligaments, the strength of the extensor muscles, the alignment of the vertebral bodies with

the adjacent vertebrae, and the compressibility of the posterior portions of the intervertebral discs. In addition to the factors listed for flexion and extension, trunk lateral movement is controlled by the impingement of each vertebra's transverse process on the adjacent transverse processes. Trunk rotation is limited by the stiffness of spinal ligaments, the strength of the muscles on the side of rotation, the stiffness of the muscles opposite the side of rotation, and body tissues and their dimensions. For instance, rotating to the left is limited by weak left-side muscles and tight right-side muscles.

Many people who have stiff back muscles have discovered that stretching helps relieve some of the pain. The back muscles, or trunk extensors, are not the only lower-trunk muscles to influence back pain. Often people find relief from back pain by leaning back (trunk hyperextension) because this action stretches the abdominal muscles, the trunk flexors. This shows that flexible trunk flexors are also important. Moreover, numerous sporting activities such as golf, tennis, and throwing sports require twisting of the trunk. Twisting the trunk involves the trunk extensors, flexors, and lateral flexors. Improved range of motion of all lower-trunk muscles can increase the range of motion in trunk rotation and improve the performance in activities that involve these actions.

Hyperextension (arching) and hyperflexion (bending) of the lower back are potentially dangerous, especially if you have weak abdominal, thigh, and buttocks muscles. Backward rolling movements are potentially dangerous to the cervical spine (neck). Potential injuries include excessively squeezing the spinal discs, jamming together the spinal joints, and pinching the spinal nerves emerging from the lumbar vertebrae. If you choose to perform these stretches, build up to them more gradually than for most other stretches. Also, to keep pressure off the neck during back rolls, keep the shoulder blades in contact with the floor.

BEGINNER STANDING HIP FLEXOR STRETCH



Execution

1. Stand upright with the left leg next to a cushioned couch or armchair, with the height of the couch cushions either even with or just a little below the knee.
2. Place the back of an additional chair facing you for support.
3. Keeping the right leg straight, bend the left knee to about 90 degrees, and rest it on the couch or armchair.
4. Grasp the back of the chair in front of you to maintain balance.
5. While holding on to the chair, slowly move the hips forward as one unit.
6. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right quadriceps (rectus femoris, vastus medialis, vastus intermedius, vastus lateralis), right iliopsoas, right tensor fasciae latae, middle and upper right sartorius

Less-stretched muscles: Right pectineus, right anterior gluteus medius, right adductor brevis, right adductor longus

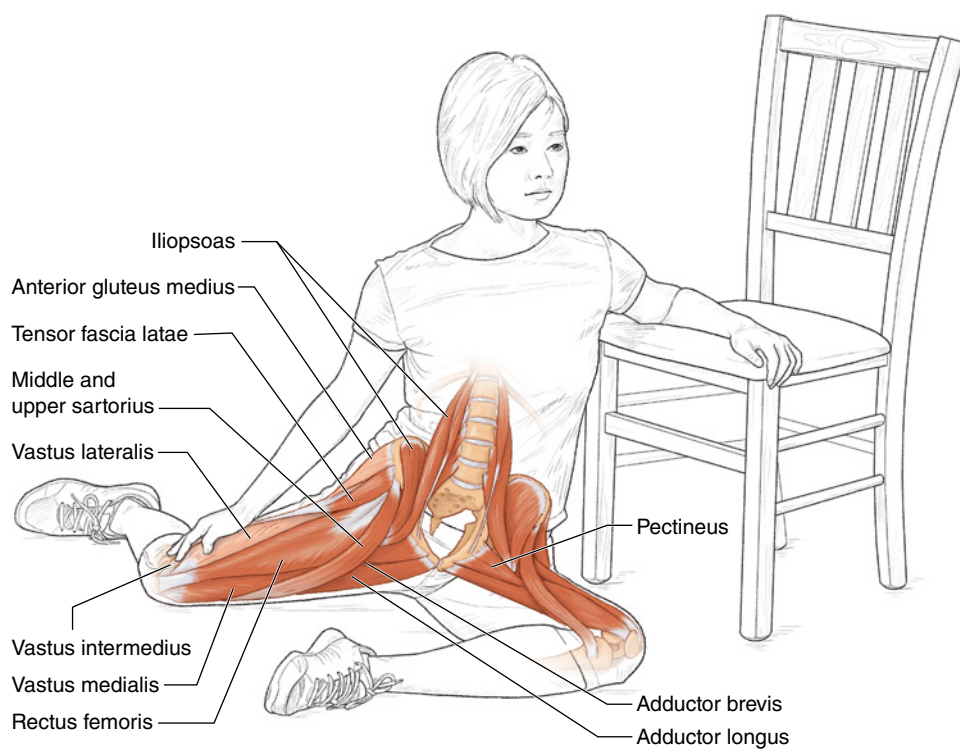
Stretch Notes

People often encounter periodic muscle soreness or tightness in front of the hip area because of certain types of exercise movements, especially during or after performing new movement patterns in exercise routines or sports. Soreness also easily occurs after restarting an exercise or sport after many months or years away from it. If muscle soreness intensifies during the performance of a sport or exercise, it is time to take a break and stretch the affected muscle groups.

When placing the bent leg on the supporting surface, use extra pillows if needed for correct height positioning, and make sure the entire lower part of that leg is resting on the cushions. This minimizes stress on the lower leg and knee joint while allowing more effective stretching of the hip flexor muscles. Also, placing extra cushioning under the bent leg will make this stretch more comfortable.

Moving the hips slowly forward as a unit while holding on to the chair in front of you gives more stretch. As your flexibility improves, you might try to arch your lower back carefully while moving your hips forward at the same time to increase the effectiveness of this stretch. In addition, rotating your torso slightly outward or inward emphasizes the stretch to the medial (inside) or lateral (outside) area of the thigh, respectively.

INTERMEDIATE HIP FLEXOR STRETCH



Execution

1. Sit on the floor near a couch, bed, or armchair.
2. Place the left leg in front of the torso, with the left knee bent to less than 90 degrees and the lateral side of the left leg flat on the floor surface.
3. Extend the right leg behind the torso, with the knee bent and the medial side of the knee and lower leg lying on the floor as far back as comfortably possible.
4. Lean the upper left torso against the couch, bed, or armchair, with the left arm lying on the surface for extra support and balance.
5. Keeping the torso upright and next to the couch, bed, or armchair, move the hips as a unit slowly forward.
6. Simultaneously, try to straighten the right knee and slide the right leg as far back as possible.
7. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right iliopsoas, right pectineus, right tensor fasciae latae, right rectus femoris, right vastus medialis, right adductor brevis, right adductor longus, middle and upper right sartorius

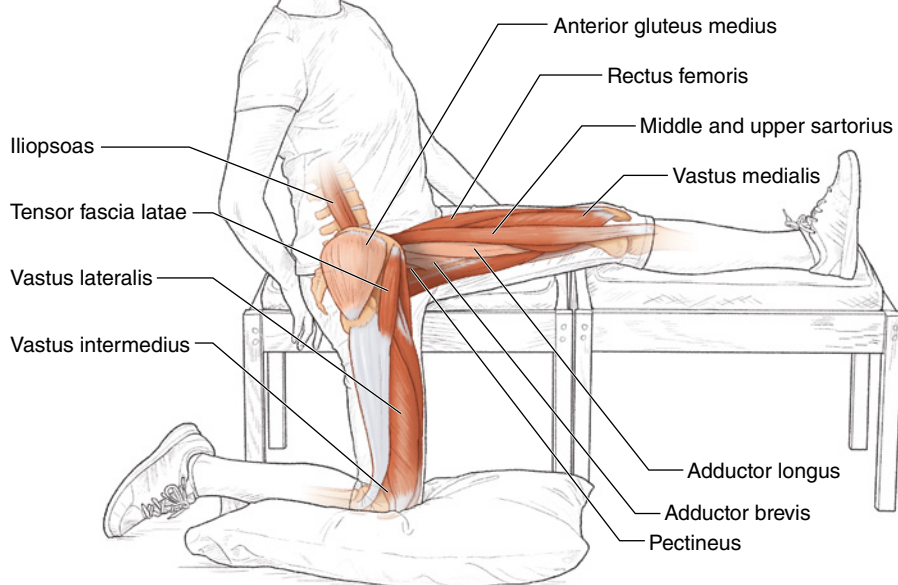
Less-stretched muscles: Right anterior gluteus medius, right vastus intermedius, right vastus lateralis

Stretch Notes

This is an excellent iliopsoas (iliacus and psoas major) stretch. It might be hard for some people (especially men) to perform this stretch at first, but after a week or so, it will gradually get easier. Since the female pelvis is shorter, larger and broader than the male pelvis, this stretch position is relatively easier for women than for men. The stretch is greater if you move the hips slowly forward as a unit while keeping the torso next to the couch, bed, or armchair and keeping the torso upright as much as possible. As your flexibility improves, you might try to arch your lower back carefully while moving your hips forward at the same time to increase the effectiveness of this stretch. Rotating your torso in a slightly inside direction will also stretch the iliopsoas muscle more.

The ease of performing this stretch will allow you to do it several times a day, especially because it can be done while reading, watching TV, or just relaxing.

ADVANCED HIP FLEXOR STRETCH



Execution

1. Sit along the edge of a couch or bed, with the left leg extended on the surface in front of you.
2. Place a pillow or cushion on the floor next to the couch or bed.
3. Bend the right knee, and rest the right knee and shin on the pillow or cushion.
4. Place the hands on the couch or bed behind the hips and torso.
5. Slowly extend the torso backward from the hip as much as is comfortable by supporting the weight of the torso with the arms and hands on the surface behind the hips.
6. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right iliopsoas, right tensor fasciae latae, right quadriceps (rectus femoris, vastus medialis, vastus intermedius, vastus lateralis), middle and upper right sartorius

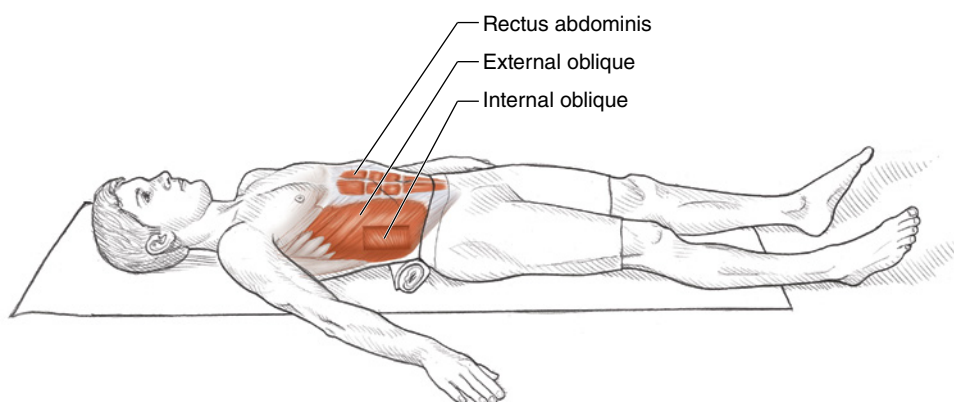
Less-stretched muscles: Right pectineus, right anterior gluteus medius, right adductor brevis right adductor longus

Stretch Notes

This is an advanced stretch for the hip flexor muscles. You can perform this stretch while sitting on the sofa or bed, while watching TV, or while listening to music. Sitting on a soft surface helps make the execution of this stretch more comfortable and relaxing. Place a pillow under the right knee and dorsal part of the foot for comfort. Extending the right leg back from the torso allows you to monitor the extent of the stretch. As flexibility increases, lean farther back, and support the weight on the elbows instead of the hands. Also, as your flexibility increases, you will find that you are able to extend the right leg farther back. In addition, moving your hips slowly forward allows you to monitor the amount of stretch on the hip flexor muscles. The intensity of this stretch can be increased by using the combination of moving the torso backward, extending the right leg farther back, and moving the hips forward.

If you experience soreness or tightness of either the lateral (outer) or medial (inner) side of the front thigh, consider emphasizing the stretch on the medial muscles (iliopsoas, vastus medialis, and pectineus) by rotating the upper body away from the medial muscles (rotate the right side clockwise) when bending backward. To place most of the stretch emphasis on the lateral muscles (vastus lateralis and tensor fasciae latae), rotate the upper body away from the lateral muscles (rotate the right side counterclockwise) when bending backward.

SUPINE LOWER-TRUNK FLEXOR STRETCH



Execution

1. Lie on the floor on your back.
2. Place a rolled-up towel (1 to 2 inches, [2.5 to 5 cm] in diameter) between the small of your back and the floor.

Muscles Stretched

Most-stretched muscles: Rectus abdominis, external oblique, internal oblique

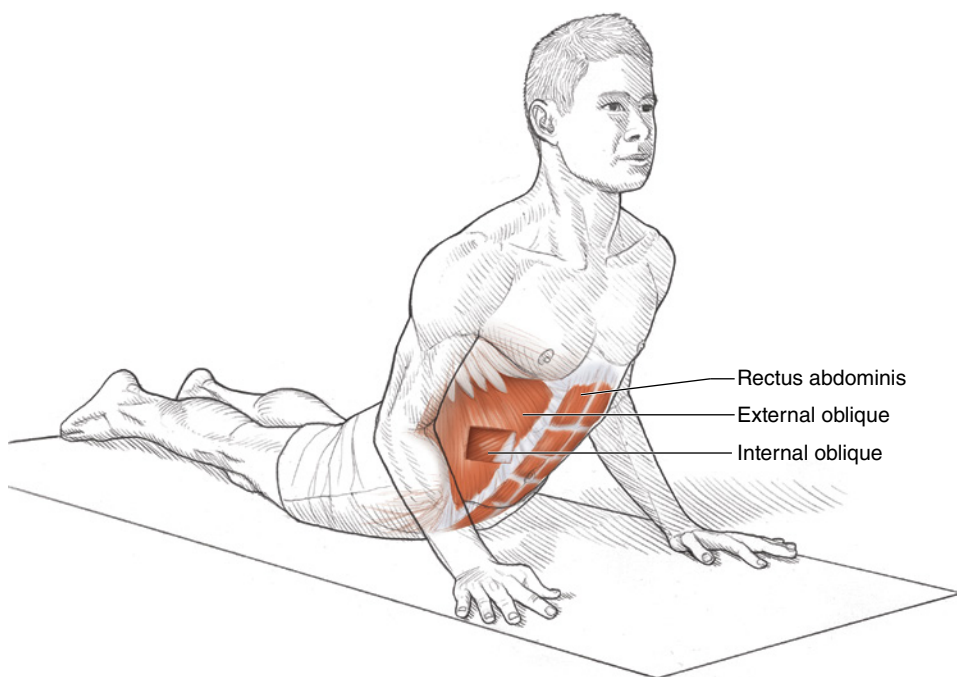
Less-stretched muscles: Quadratus lumborum, psoas major, iliacus

Stretch Notes

Although many people think that having tight abdominal muscles improves their overall appearance, tight abdominal muscles can have very negative effects on the body. First, tight abdominals are a major cause of lower-back pain. When tight, these muscles pull up on the pubic bones and tilt the top of the pelvis backward. Over time, the upper-back muscles weaken and overlengthen, causing a flattening of the lumbar curve, which increases the pressure on the lumbar joints and discs. The constant stretch and compression of the discs result in chronic pain. In addition, when these muscles are tight, the volume of the abdominal and pelvic cavities is reduced. This compresses the organs in these cavities and forces them up toward the thoracic cavity, which in turn reduces its volume. As a result, breathing, digestion, elimination, and sexual function are hindered from functioning properly. Finally, exercising with tight abdominals can lead to strains, tears, and even hernias.

This stretch is especially recommended for people who have a swayed back or weak abdominal muscles, because arching the lower back is potentially dangerous for these people. Because the small of the back is supported in this exercise, undesired pressures on the spinal column are reduced. Nevertheless, the width of the back support is important. The larger the diameter of the towel, the greater the undesired pressure. Make sure the upper back, shoulder blades, and buttocks are resting comfortably on the floor. Also, squeezing the buttocks will reduce stress on the lower back.

PRONE LOWER-TRUNK FLEXOR STRETCH



Execution

1. Lie prone (facedown) on the floor.
2. Place both hands palms down. Fingers point forward by each hip.
3. Slowly arch the back, contracting the buttocks.
4. Continue arching the back as you lift your head and chest off the floor without hunching the shoulders.

Muscles Stretched

Most-stretched muscles: Rectus abdominis, external oblique, internal oblique

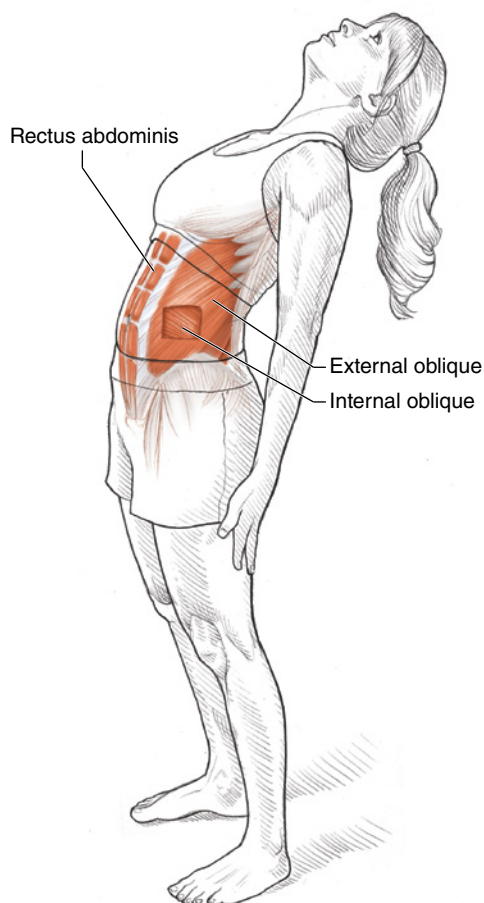
Less-stretched muscles: Quadratus lumborum, psoas major, iliacus, rotatores, intertransversarii

Stretch Notes

Those who spend a lot of time driving or sitting at a desk tend to slump forward, rounding the upper back, which also tightens the abdominal muscles. Having tight abdominal muscles is equivalent to wearing a corset. This compression of the abdominal and pelvic cavities can cause deterioration of the back muscles, restrict breathing, and interfere with the working of other visceral organs (e.g. stomach, intestines, liver, pancreas). When these muscles are tight, the diaphragm cannot lower and the rib cage cannot expand. Poor respiration can result in chronic fatigue, depression, asthma, and other consequences caused by inadequate oxygenation of the blood. Also, the organs in the abdominal cavity do not work well in a confined space. Kidney and bladder functions may be reduced. The uterus can be forced downward, increasing pressure and reducing blood flow. Increased pressure and decreased blood flow to the prostate may occur.

Remember that arching the lower back is potentially dangerous, especially if you have weak abdominal muscles. Injuries from arching the lower back include excessive squeezing of the spinal discs, jammed spinal joints, and pinched spinal nerves emerging from the lumbar vertebrae. Therefore, this stretch is recommended only for those who are very stiff. When doing this stretch, arch minimally and make sure you squeeze the buttocks during the arching. Squeezing the buttocks reduces stress on the lower back.

STANDING LOWER-TRUNK FLEXOR STRETCH



Execution

1. Stand upright with the legs 2 to 3 feet (60 to 90 cm) apart, with hands on the hips.
2. Slowly arch the back, contracting the buttocks and pushing the hips forward.
3. As you continue to arch the back, drop the head back and slide the hands past the buttocks and down the legs.

Muscles Stretched

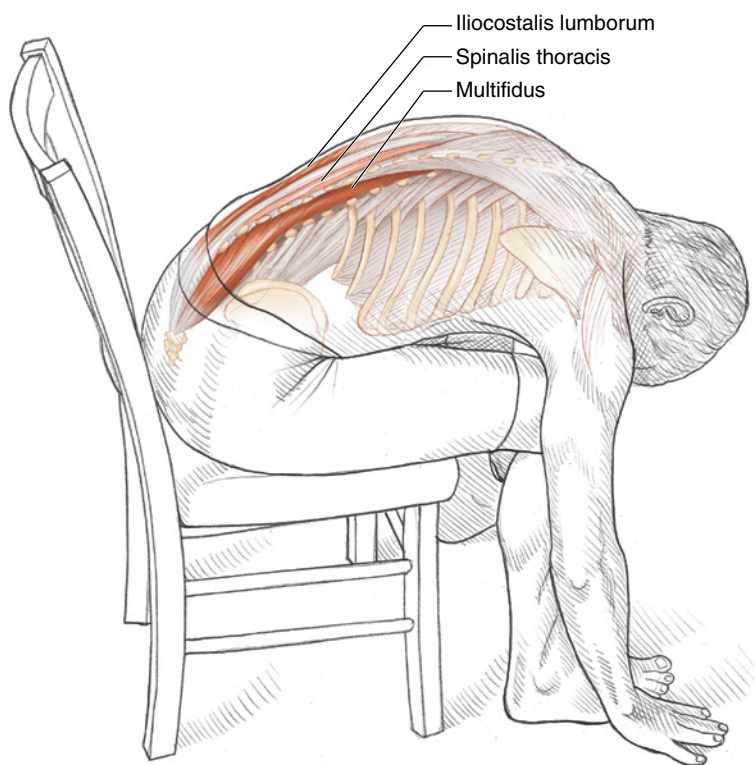
Most-stretched muscles: Rectus abdominis, external oblique, internal oblique

Less-stretched muscles: Quadratus lumborum, psoas major, iliacus

Stretch Notes

This exercise is potentially dangerous, especially for those who have a swayed back or weak abdominal muscles. This exercise can worsen a swayed back and cause excessive squeezing of the spinal discs, jammed spinal joints, and pinched spinal nerves emerging from the lumbar vertebrae. This stretch is recommended only for those who are very stiff and do not have a swayed back. Also, use this exercise only when the other lower-back flexor stretches do not provide improvement. When doing this stretch, arch minimally, and make sure you squeeze the buttocks during the arching. Squeezing the buttocks reduces the stress on the lower back.

SEATED LOWER-TRUNK EXTENSOR STRETCH



Execution

1. Sit upright in a chair, with legs separated.
2. Slowly round the upper back and begin to lean forward.
3. Continue to bend at the waist and lower the head and abdomen between the legs and below the thighs.

Muscles Stretched

Most-stretched muscles: Iliocostalis lumborum, multifidus

Less-stretched muscles: Interspinales, rotatores, spinalis thoracis

Stretch Notes

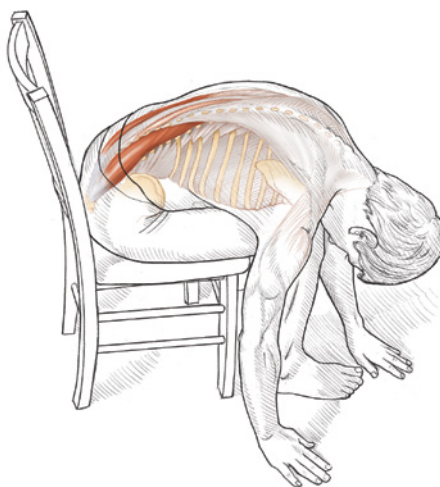
When done with incorrect posture, simple daily tasks such as housecleaning, gardening, lifting heavy objects, and exercising can cause tightness in back muscles. Poor posture includes slouching in chairs, sitting in hunched-back positions, standing in nonupright positions, and keeping the knees straight when lifting. All these actions lead to tight muscles by either overworking or overstretching the back muscles. Two other common reasons for tight back muscles are conscious psychological stress and subconscious repressed emotions. Stress causes back muscles to tighten in a fight or flight response, thus overworking the muscles and depriving them of the energy needed to support the spine. In the short term, back-stretching exercises reduce these problems by reducing stress. In the long term, these exercises make the back muscles stronger and longer and thus reduce the possibilities of overworking and overstretching.

Remember that hyperflexion can injure the spinal cord. When doing this exercise, go slowly and do not let the back become straight. Also, the effect of the stretch is minimized if the buttocks rise off the chair.

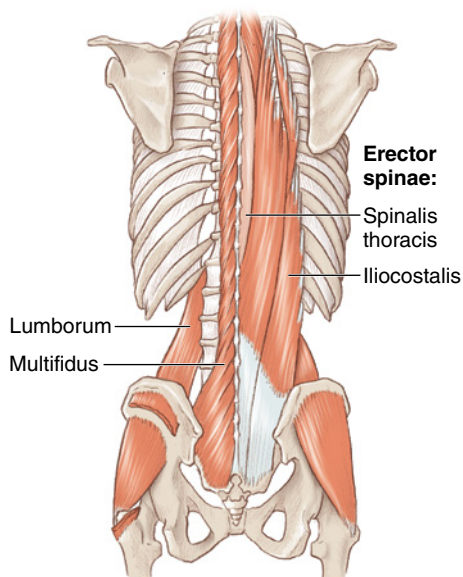
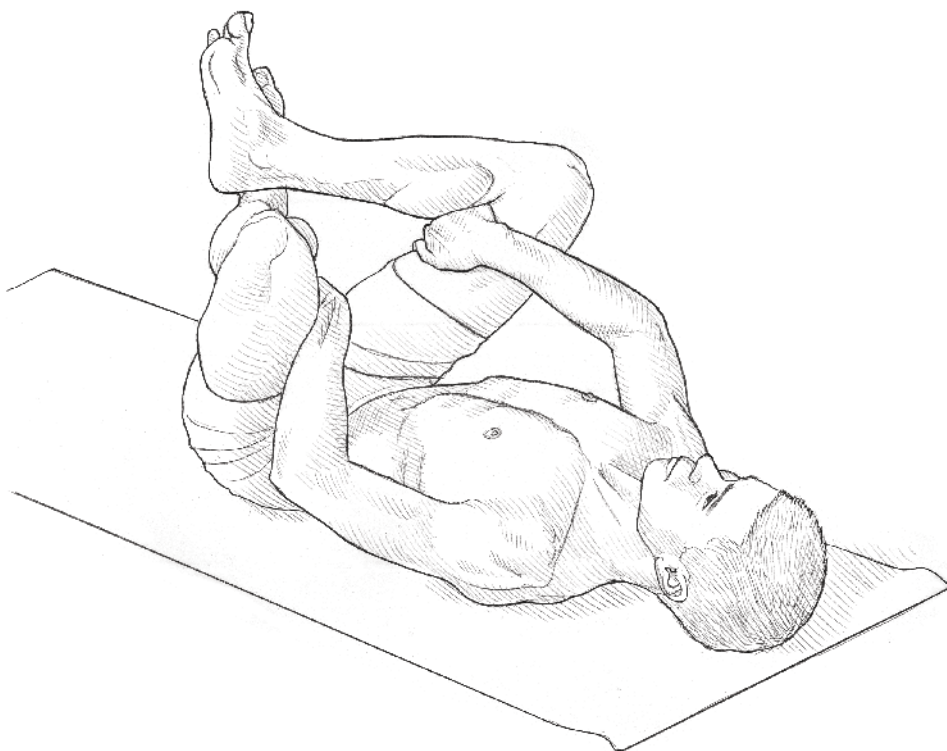
VARIATION

Seated Lower-Trunk Extensor Lateral Flexor Stretch

Angling the head toward one of the knees will increase the stretch on the lower-trunk extensors and partially stretch some of the lateral flexors. Sit upright in a chair, with legs separated. Slowly extend the upper back and lean forward. As you lean forward, continue to bend at the waist and lower your head and abdomen toward the right knee. Finally, slowly lower the head below the right knee. Repeat toward the left knee.



RECLINING LOWER-TRUNK EXTENSOR STRETCH



Execution

1. Lie on the back with the legs extended.
2. Flex the knees and hips, bringing the knees to the chest.
3. Cross the feet at the ankles, and separate the knees so they are at least shoulder-width apart.
4. Grasp the thighs inside of the knees, and pull the legs down to the chest.

Muscles Stretched

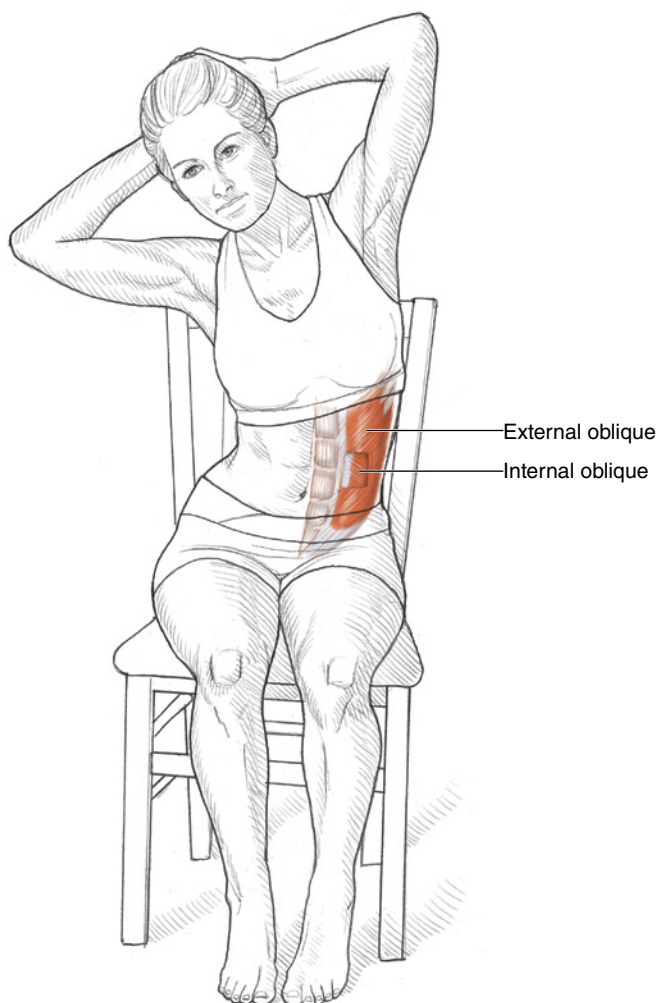
Most-stretched muscles: Iliocostalis lumborum, multifidus

Less-stretched muscles: Interspinales, rotatores, spinalis thoracis

Stretch Notes

Some people find that when they are performing the seated lower-trunk extensor stretch, they cannot lean forward slowly without contracting the back muscles. Keeping the muscles contracted while performing a stretch greatly reduces the effect of the stretch. Because the legs can weigh less than the trunk, these people may find it easier to perform this stretch while reclining. Also, because hyperflexion can injure the spinal cord, this stretch may be safer than the seated lower-trunk extensor stretch. When doing the reclining lower-trunk extensor stretch, it is easier to go slowly and not let the back straighten. By bringing the legs to the chest, you can easily raise the buttocks off the floor and prevent a straight back by allowing the spinal column to curl. Finally, do not try to bring the knees too far below the chest (do not try to touch the knees to the floor) because this could negate the safety benefits of this stretch.

BEGINNER LOWER-TRUNK LATERAL FLEXOR STRETCH



Execution

1. Sit upright in a chair.
2. Interlock the hands behind the head, with the elbows in a straight line across the shoulders.
3. While keeping both elbows back and in a straight line, laterally flex the waist and move the right elbow toward the right hip.
4. Repeat these steps for the opposite side.

Muscles Stretched

Most-stretched muscles: Left external oblique, left internal oblique, left rotatores

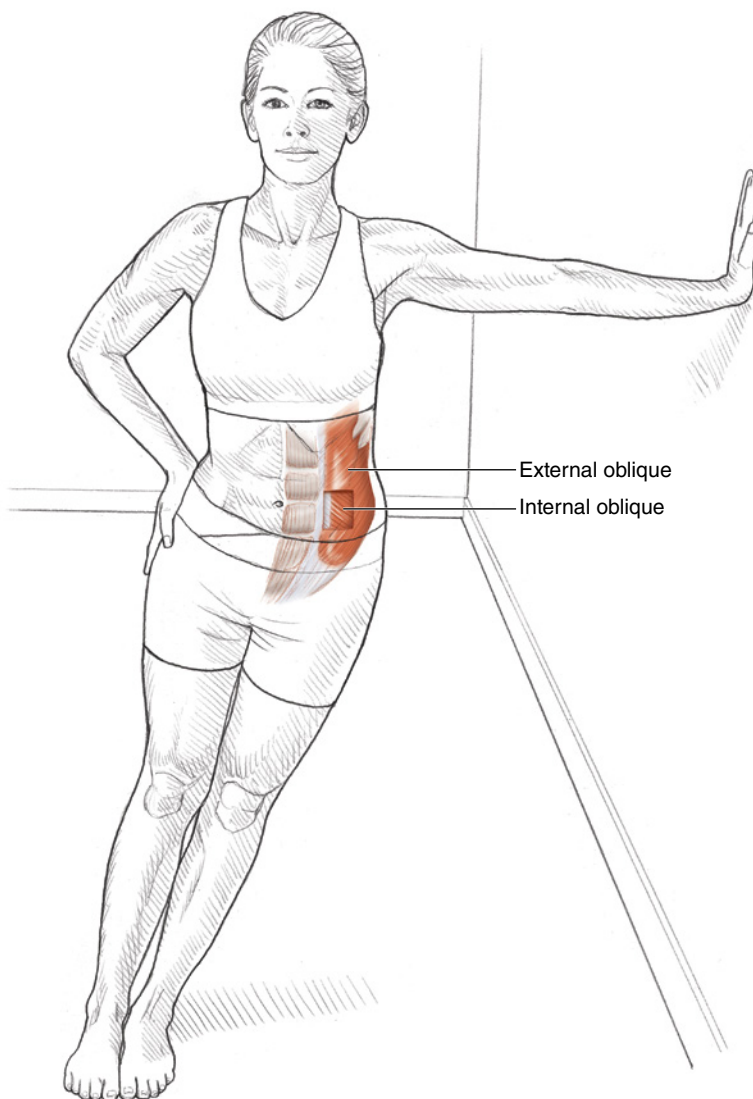
Less-stretched muscles: Left intertransversarii, left multifidus, left quadratus lumborum

Stretch Notes

Research has shown that the inability to achieve lateral flexion is a risk indicator for recurrent nonspecific low-back pain and injuries. Also, athletes who perform overhead actions for maximum distance or force, such as baseball players, football quarterbacks, and javelin throwers, need loose lateral flexors. They are also important for overhead hitting (e.g., racket-sport serves and smashes) and when reaching as high as possible (e.g., rebounding a basketball or spiking a volleyball). Gymnasts, modern and ballet dancers, and divers need these muscles to be loose. In addition, tight lateral flexors can lead to a form of scoliosis. The quadratus lumborum's only action is lateral flexion, and tightness in this muscle results in a loss of lateral stability of the spine, causing the spine to curve to the left or right.

Flexing or extending at the waist will reduce this stretch's effectiveness. Also, keep the buttocks and thighs in complete contact with the chair. The closer the elbow gets to the floor, the harder it will be to remain seated in the chair. Wrapping the lower legs and feet around the chair legs will help keep the buttocks and thighs in contact with the seat.

INTERMEDIATE LOWER-TRUNK LATERAL FLEXOR STRETCH



Execution

1. Stand upright with the feet together and the left side of the body facing a wall about an arm's length away.
2. Place the palm of the left hand on the wall at shoulder height. Place the heel of the right hand at the hip joint.
3. While keeping the legs straight, contract the buttocks and slightly rotate the hips in toward the wall.
4. Use the right hand to push the right hip toward the wall.
5. Repeat these steps for the opposite side.

Muscles Stretched

Most-stretched muscles: Left external oblique, left internal oblique, left rotatores

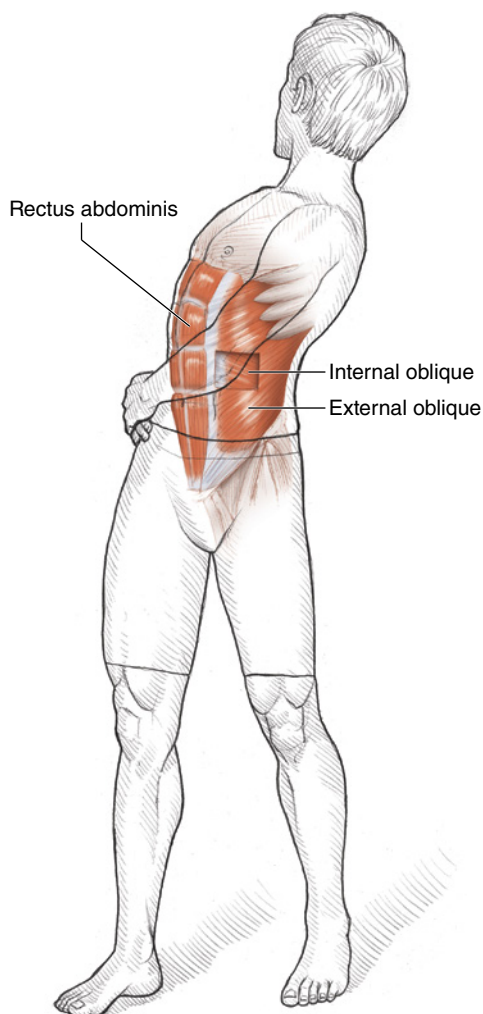
Less-stretched muscles: Left intertransversarii, left multifidus, left quadratus lumborum

Stretch Notes

Many sports rely on lateral trunk flexion. Because many of these activities stress one side of the body more than the other, it is easy for the two sides of the body to become unbalanced. The active side can become tight from being overworked. If the nonworking side goes unused for extended periods, the muscles can become short. Imbalances between the sides of the body can also result from heavy lifting, especially if one side is substantially stronger, or from participation in activities such as martial arts and football in which the body receives heavy blows. This exercise is better suited than the basic lower-trunk lateral flexor stretch for restoring flexibility because the person is in a standing position similar to that used in a variety of sports activities.

It is easy to lose balance while doing this exercise, so stand on a nonskid surface. Keep the left arm straight, but do not lock the elbow. You can increase the amount of stretch by moving the feet farther from the wall, by resting the left forearm instead of the hand on the wall, or both.

ADVANCED STANDING LOWER-TRUNK LATERAL FLEXOR STRETCH



Execution

1. Stand upright with legs 2 to 3 feet apart (60 to 90 cm), with the right foot about 1 foot (30 cm) ahead of the left foot.
2. Place both hands near the right hip.
3. Slowly arch the back, contracting the buttocks and pushing the hips forward.
4. As you continue to arch the back, rotate the trunk to the left and drop the head back toward the right side.
5. Slide the hands past the right buttock and down the right leg.
6. Repeat these steps for the opposite side.

Muscles Stretched

Most-stretched muscles: Rectus abdominis, left external oblique, left internal oblique

Less-stretched muscles: Left quadratus lumborum, left psoas major, left iliacus, left rotatores, left intertransversarii

Stretch Notes

This exercise is potentially dangerous, especially for people with a swayed back or weak abdominal muscles. This exercise can worsen a swayed back and cause excessive squeezing of the spinal discs, jammed spinal joints, and pinched spinal nerves emerging from the lumbar vertebrae. This stretch is recommended only for those who are very stiff and do not have a swayed back. Also, you should use this exercise only when the other lower-back flexor stretches do not provide improvement. When doing this stretch, arch minimally, and make sure you squeeze the buttocks during the arching. Squeezing the buttocks reduces the stress on the lower back. Finally, it is easy to lose balance while doing this exercise, so take extra care.

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